

Department of Physical Science
Faculty of Applied Science
University of Vavuniya
IT 3152 – Software Quality Assurance
Assignment 03

Date of Deadline: 13th OCT 2025

1.
 - a) Briefly explain the concept of **Software Quality Assurance (SQA)**.
 - b) List down the **seven key attributes** of **Software Quality Attributes**.
 - c) **Compare and contrast** between **Software Quality Assurance** and **Software Testing**.
 - d) Briefly explain the **factors that help to achieve software quality**.
 - e) Briefly explain the **role of the SQA team** in the **software development process**.
 - f) Explain the **three categories of McCall's Quality Model** and give a **real-world example** for each category.

2.
 - a) **List down five key activities in contract review of pre-project software quality components.**
 - b) **Briefly explain the main issues in the project development plan during contract review.**
 - c) **List down the main stages involved in the QA process and briefly explain each stage.**
 - d) **Compare and contrast regression testing and release testing in the Quality Assurance process.**
 - e) During testing of an **e-commerce website**, the tester notices that the **search results page is blank** when a user searches for a product. The tester logs this as a **New bug**, and it goes through the following stages:
 - **New:** The tester reports the issue with steps and details.
 - **Assigned:** The defect is assigned to a developer for investigation.
 - **Open:** The developer starts debugging and identifies the issue in the backend code.
 - **Fixed:** The developer corrects the issue and marks it as fixed.
 - **Pending Retest:** The tester is notified to recheck the functionality.
 - **Retest:** The tester verifies the fix and confirms that the search results now appear correctly.
 - **Verified:** The tester confirms that the defect is resolved.
 - **Closed:** The issue is closed after successful verification.
 - i. **Explain how the bug in the e-commerce website progresses through different states of the Bug Life Cycle.**
 - ii. **Why is it important for testers and developers to communicate effectively during each stage of the Bug Life Cycle in this scenario?**

3.

- a. **Briefly explain the importance of software testing.**
- b. **Compare and contrast White-box testing, Black-box testing, and Gray-box testing.**
- c. **List down the types of test cases in Software Quality Assurance.**
- d. **Briefly explain each of the components of a test case:**
 - i. Test Steps
 - ii. Test Data
 - iii. Expected Results
 - iv. Status
 - v. Post Conditions
- e. **Design test cases for Gmail login functionality and write at least 4 test cases.**

4.

A company is launching a new **online food delivery app** with several features, including **user registration and login, browsing restaurants and menus, adding items to a cart, payment through credit card, debit card, and wallet, and tracking order status** (Placed → Prepared → On the way → Delivered). The app needs thorough testing to ensure all features work correctly and provide a smooth user experience.

- a) **Test Design Activities:** List the four key activities of test design you would carry out for this app (test case design, test data design, test environment design, test procedure design) and give **one example for each**.
- b) **Test Design Techniques:** Using **Equivalence Partitioning**, identify **three input partitions** for the “credit card number” field in the payment feature.
- c) **Boundary Value Analysis:** The “Promo Code” field accepts codes between **5 and 10 characters**. Write the **test cases** you would create for boundary value analysis.
- d) **State Transition Testing:** Draw the **state transition diagram** for the “Order Status” feature (Placed → Prepared → On the way → Delivered) and mention **one valid** and **one invalid transition** you would test.
- e) **Risk & Coverage:** Explain how **test design** in this scenario would help identify **critical/risky areas** of the app (like payments) and ensure **full test coverage** before release.